DATE PAGE

DEEP LEARNING
ASSIGNMENT - 1

(6)

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Implement XOR function using MP Neuron.

x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

XOR function is represent as $y = x_1\bar{x}_2 + \bar{x}_1 x_2$

$$y = x_1\bar{x}_2 + \bar{x}_1 x_2$$

$$= z_1 + z_2$$

or

$$= z_1 \cup z_2$$

Compute the net inputs,

$w_1 = 1$, $w_2 = -1$

(0,0) z_1 in $= 0\times1 + 0\times-1 = 0$

(0,1) z_1 in $= 0\times1 + 1\times-1 = -1$

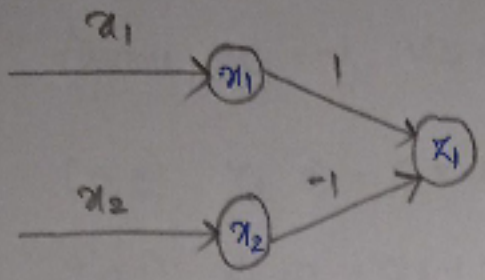
(1,0) z_1 in $= 1\times1 + 0\times-1 = 1$

(1,1) z_1 in $= 1\times1 + 1\times-1 = 0$

On the basis of this input this is possible to get the requried output as $w_{11} = 1$ and $w_{21} = -1$

Condition $\theta \geq 1$ for the z_1 neuron

$Z_1 = x_1 \bar{x}_2$



$Z_1 = x_1 \bar{x}_2$

x_1	$\bar{x}_2$	Z_1
0	1	0
1	1	1
0	0	0
1	0	0

$Z_2 = \bar{x}_1 x_2$

$\bar{x}_1$	x_2	Z_1
1	1	0
0	1	1
0	0	0
1	0	1

SHENBAGAN

Compute net input Z_2

Assume, $w_1 = -1 \quad w_2 = 1$

(0,0) Z_2 in $= 0 \times -1 + 0 \times 1 = 0$

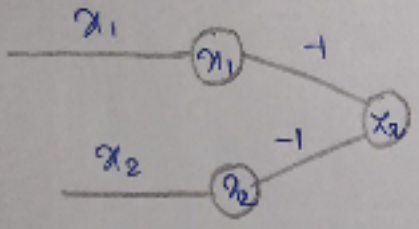
(0,1) Z_2 in $= 0 \times -1 + 1 \times 0 = 1$

(1,0) Z_2 in $= 1 \times -1 + 0 \times 1 = -1$

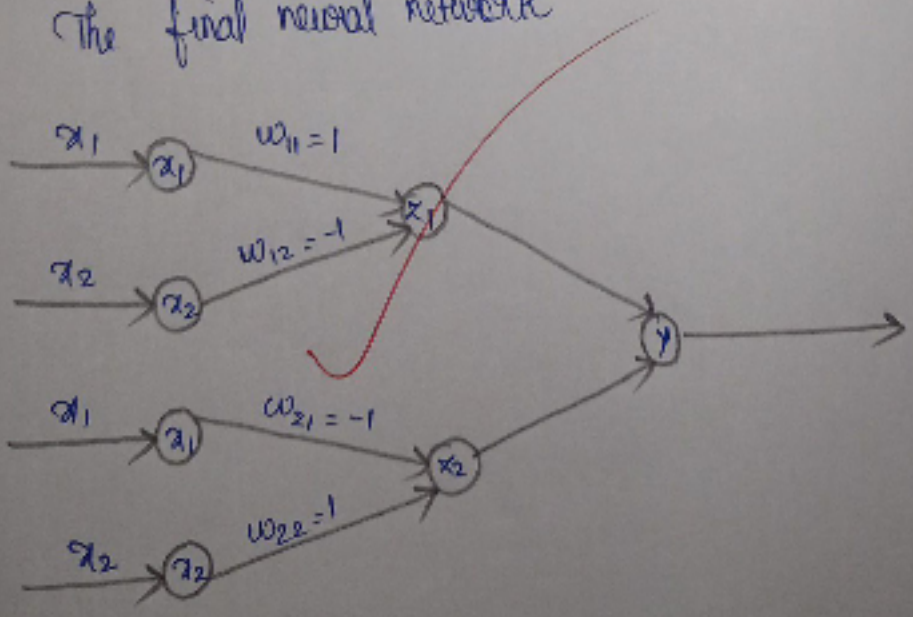
(1,1) Z_2 in $= 1 \times -1 + 1 \times 1 = 0$

On the basis of the net input it is possible to get requested output.

Hence $w_{21} = -1$, $w_{22} = 1$



The final neural network



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